

STAT4052 Final Project

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```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## Warning: package 'tibble' was built under R version 4.4.3
```

```
## Warning: package 'tidyr' was built under R version 4.4.3
```

```
## Warning: package 'readr' was built under R version 4.4.3
```

```
## Warning: package 'purrr' was built under R version 4.4.3
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
## Warning: package 'stringr' was built under R version 4.4.3
```

```
## Warning: package 'forcats' was built under R version 4.4.3
```

```
## Warning: package 'lubridate' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.1      v stringr    1.5.2
```

```
## v ggplot2    4.0.1      v tibble     3.3.0
```

```
## v lubridate  1.9.4      v tidyr      1.3.1
```

```
## v purrr      1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
```

```
library(MASS)
```

```
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select

library(car)

## Warning: package 'car' was built under R version 4.4.3
## Loading required package: carData
## Warning: package 'carData' was built under R version 4.4.3
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some

library(glmnet)

## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
## Loaded glmnet 4.1-8

library(randomForest)

## Warning: package 'randomForest' was built under R version 4.4.3
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```

library(gridExtra)

## Warning: package 'gridExtra' was built under R version 4.4.3

##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:randomForest':
##
##     combine
##
## The following object is masked from 'package:dplyr':
##
##     combine

# Data Prep
raw_data <- read_csv("C:/Users/julia/Downloads/Austin_Animal_Center_Outcomes_20260506.csv")

## Rows: 22444 Columns: 16
## -- Column specification -----
## Delimiter: ","
## chr  (12): Name, Outcome Status, Euthanasia Reason, Type, Sex, Spayed/Neuter...
## dbl  (2): ID, Animal ID
## num  (1): Days in Shelter
## dtm  (1): Outcome Date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

clean_data <- raw_data %>%
  filter(Type == "Dog") %>%
  mutate(
    Days_in_Shelter = as.numeric(gsub(",", "", `Days in Shelter`)),
    DOB = ymd_hms(`Date of Birth`),
    Intake_Date = ymd_hms(`Intake Date`),
    Age_at_Intake = as.numeric(difftime(Intake_Date, DOB, units = "days")) / 365.25,
    Intake_Month = factor(month(Intake_Date, label = TRUE)),
    sqrt_days = sqrt(Days_in_Shelter)
  ) %>%
  mutate(Breed_Cluster = case_when(
    grepl("Pit Bull", `Primary Breed`) ~ "Pit Bull Group",
    grepl("Retriever", `Primary Breed`) ~ "Retriever Group",
    grepl("Shepherd", `Primary Breed`) ~ "Shepherd Group",
    grepl("Chihuahua", `Primary Breed`) ~ "Chihuahua Group",
    grepl("Terrier", `Primary Breed`) ~ "Terrier Group",
    grepl("Hound", `Primary Breed`) ~ "Hound Group",
    grepl("Husky", `Primary Breed`) ~ "Husky Group",
    grepl("Boxer", `Primary Breed`) ~ "Boxer Group",
    TRUE ~ "Other/Small Mix"
  )) %>%
  rename(Spayed_Neuterred = "Spayed/Neuterred") %>%
  mutate(

```

```

Breed_Cluster = as.factor(Breed_Cluster),
Spayed_Neutered = as.factor(str_to_title(as.character(Spayed_Neutered))),
Sex = as.factor(Sex)
) %>%
filter(!is.na(Age_at_Intake), Age_at_Intake >= 0) %>%
dplyr::select(sqrt_days, Days_in_Shelter, Age_at_Intake, Breed_Cluster, Sex, Spayed_Neutered, Intake_L
drop_na()

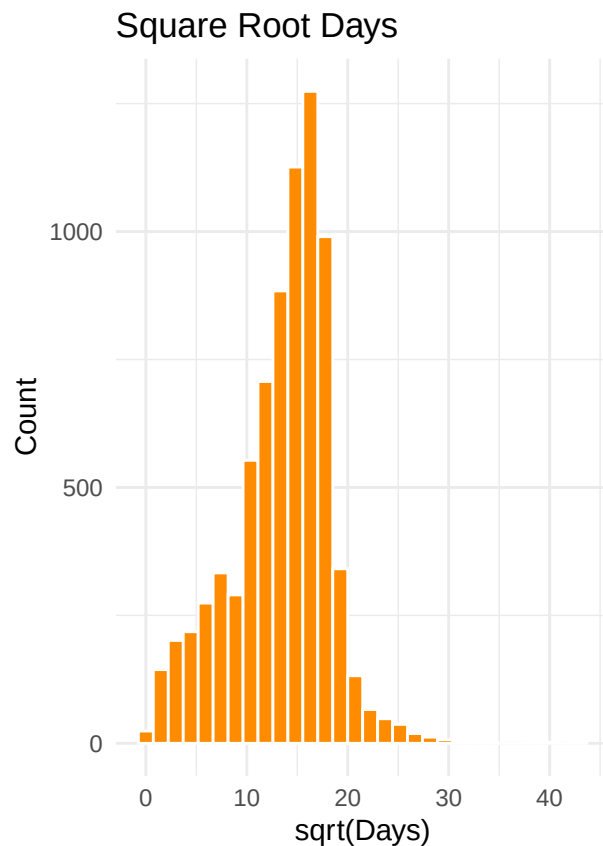
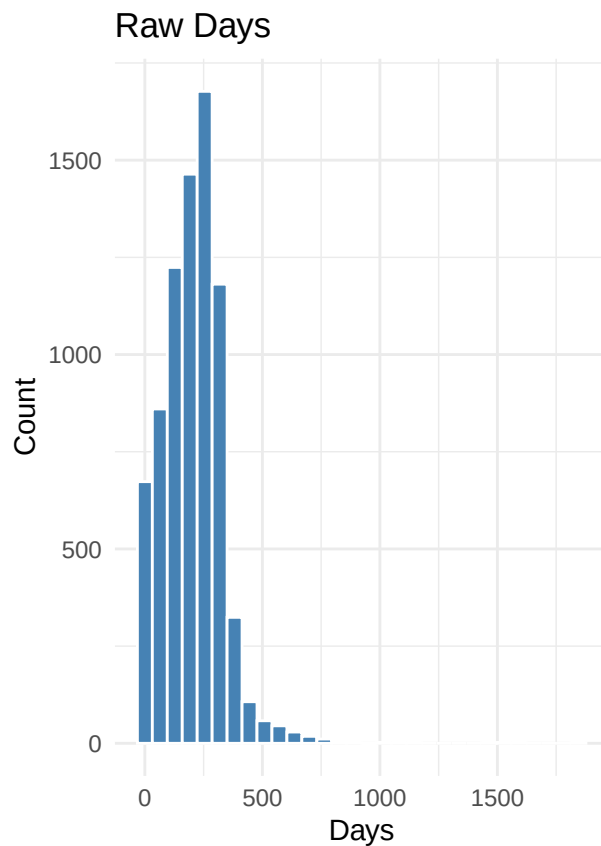
# Train Test Split
set.seed(4052)
train_indices <- sample(seq_len(nrow(clean_data)), size = floor(0.8 * nrow(clean_data)))
train_data <- clean_data[train_indices, ]
test_data <- clean_data[-train_indices, ]

# Histograms
p1 <- ggplot(clean_data, aes(x = Days_in_Shelter)) +
  geom_histogram(fill = "steelblue", color = "white", bins = 30) +
  theme_minimal() + labs(title = "Raw Days", x = "Days", y = "Count")

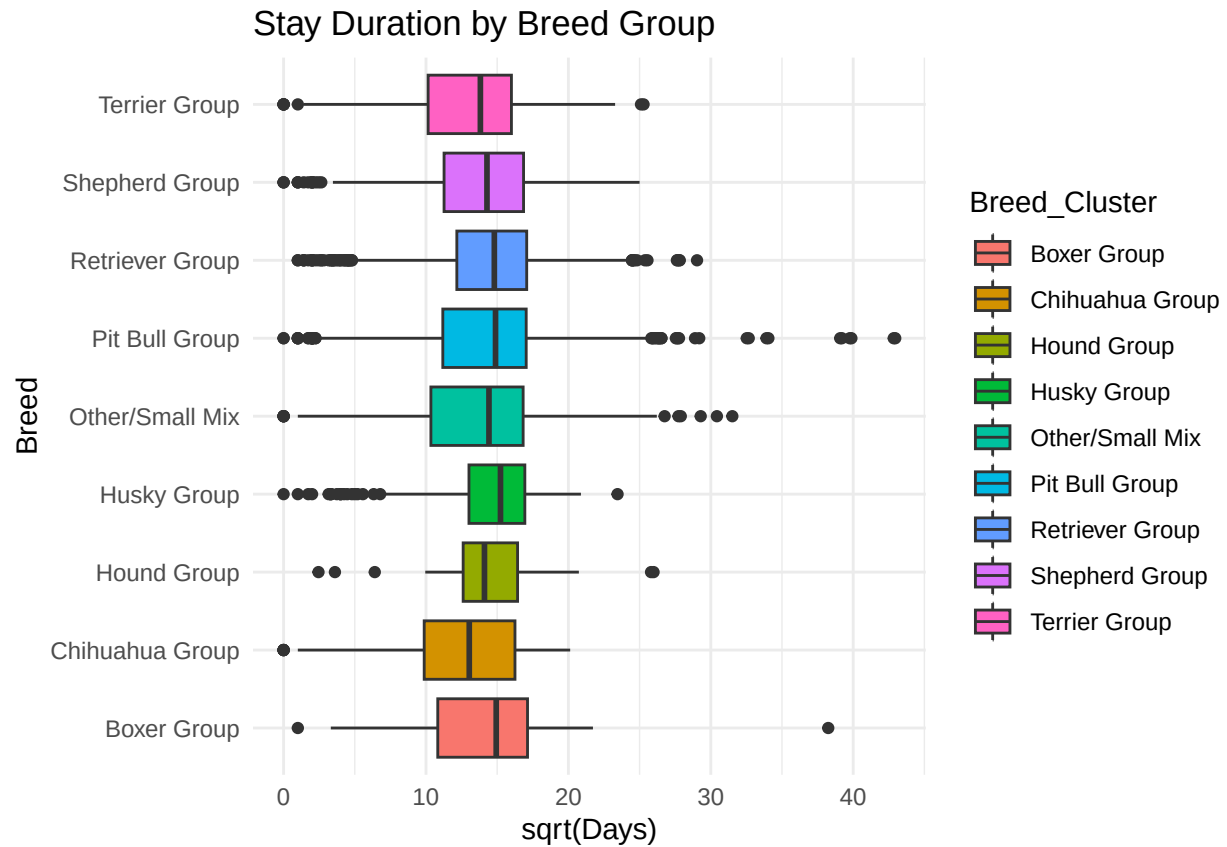
p2 <- ggplot(clean_data, aes(x = sqrt_days)) +
  geom_histogram(fill = "darkorange", color = "white", bins = 30) +
  theme_minimal() + labs(title = "Square Root Days", x = "sqrt(Days)", y = "Count")

grid.arrange(p1, p2, ncol = 2)

```



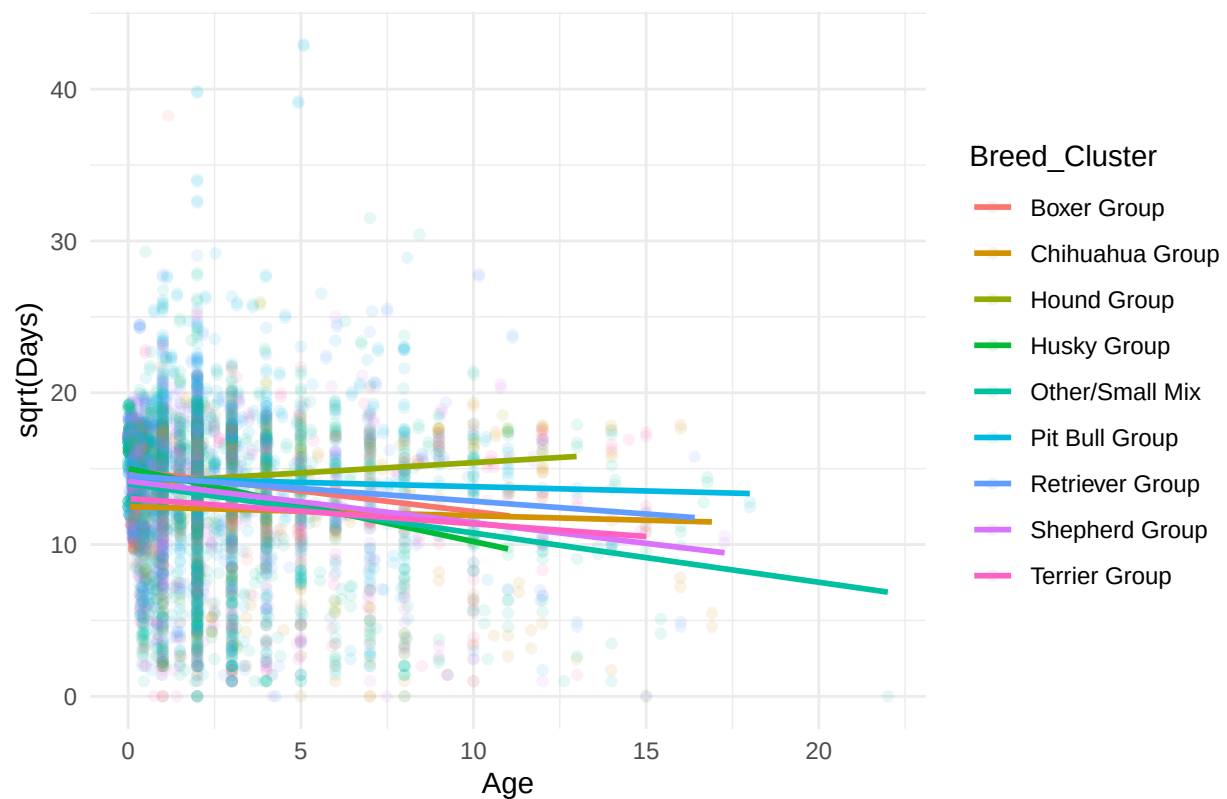
```
# Boxplot by Breed
ggplot(train_data, aes(x = Breed_Cluster, y = sqrt_days, fill = Breed_Cluster)) +
  geom_boxplot() + coord_flip() + theme_minimal() +
  labs(title = "Stay Duration by Breed Group", y = "sqrt(Days)", x = "Breed")
```



```
ggplot(train_data, aes(x = Age_at_Intake, y = sqrt_days, color = Breed_Cluster)) +
  geom_point(alpha = 0.1) + geom_smooth(method = "lm", se = FALSE) +
  theme_minimal() + labs(title = "Age and Breed Interaction", x = "Age", y = "sqrt(Days)")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Age and Breed Interaction



```
# Base vs. Enhanced MLR
```

```
# Base MLR
```

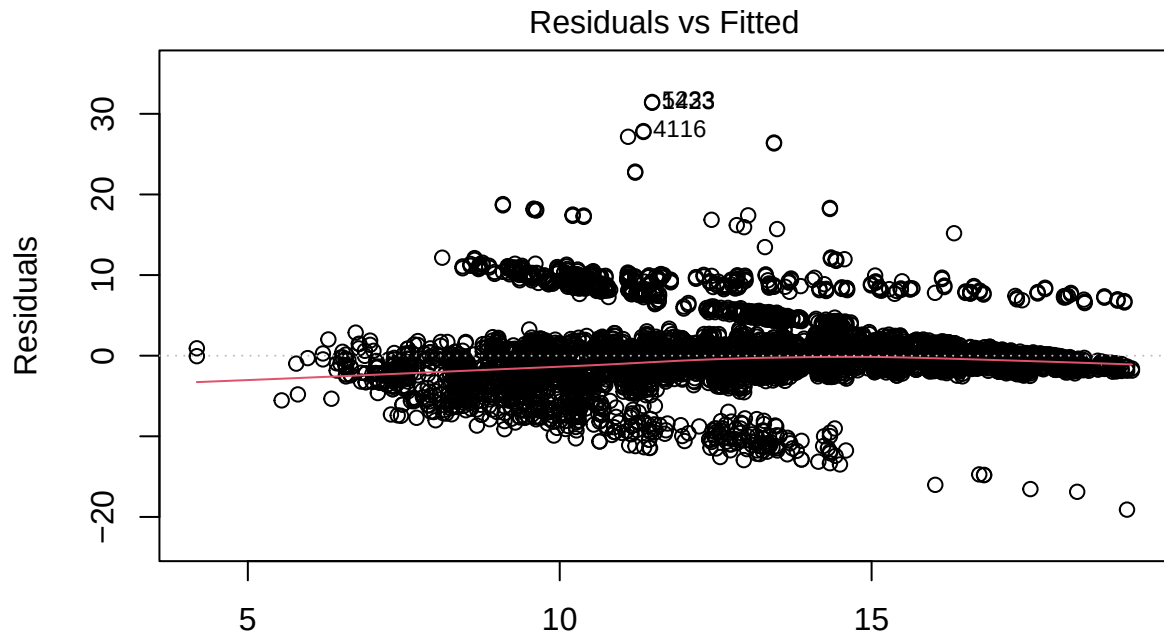
```
mlr_base <- lm(sqrt_days ~ Breed_Cluster + Sex + Spayed_Neutered + Age_at_Intake, data = train_data)
```

```
# Enhanced MLR
```

```
mlr_enhanced <- lm(sqrt_days ~ Breed_Cluster + Sex + Spayed_Neutered + Intake_Month +  
  Age_at_Intake + I(Age_at_Intake^2) + Age_at_Intake:Breed_Cluster,  
  data = train_data)
```

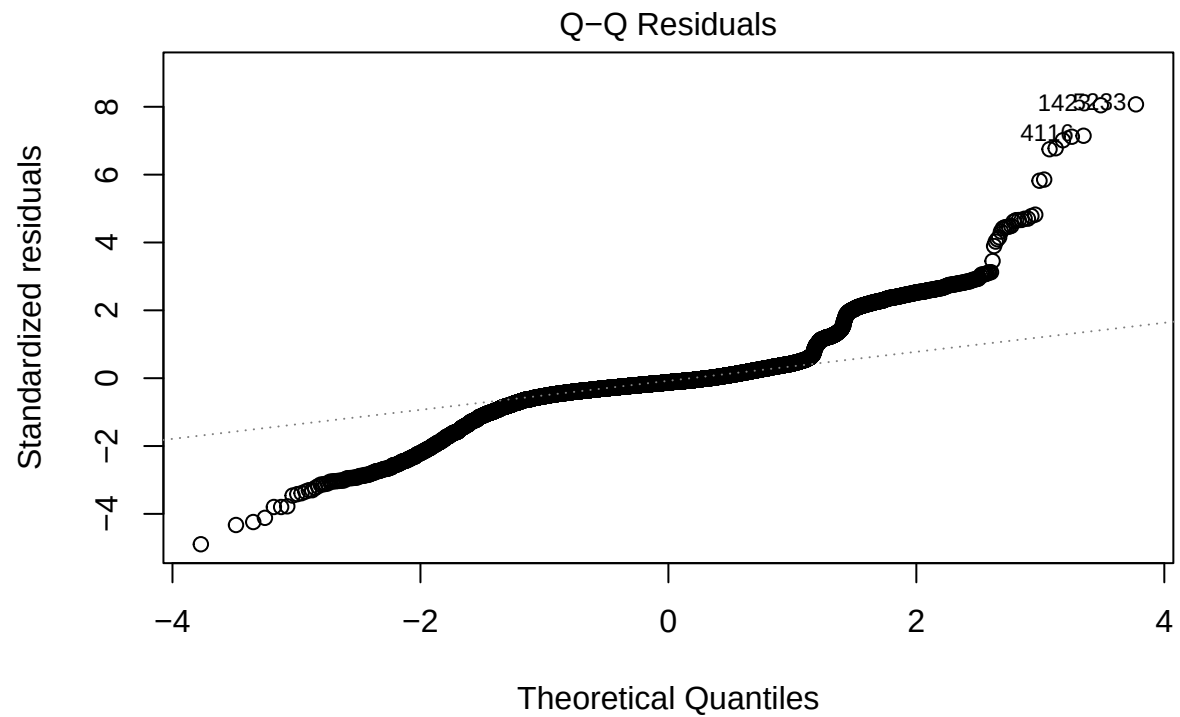
```
# Residual Plots for Enhanced Model
```

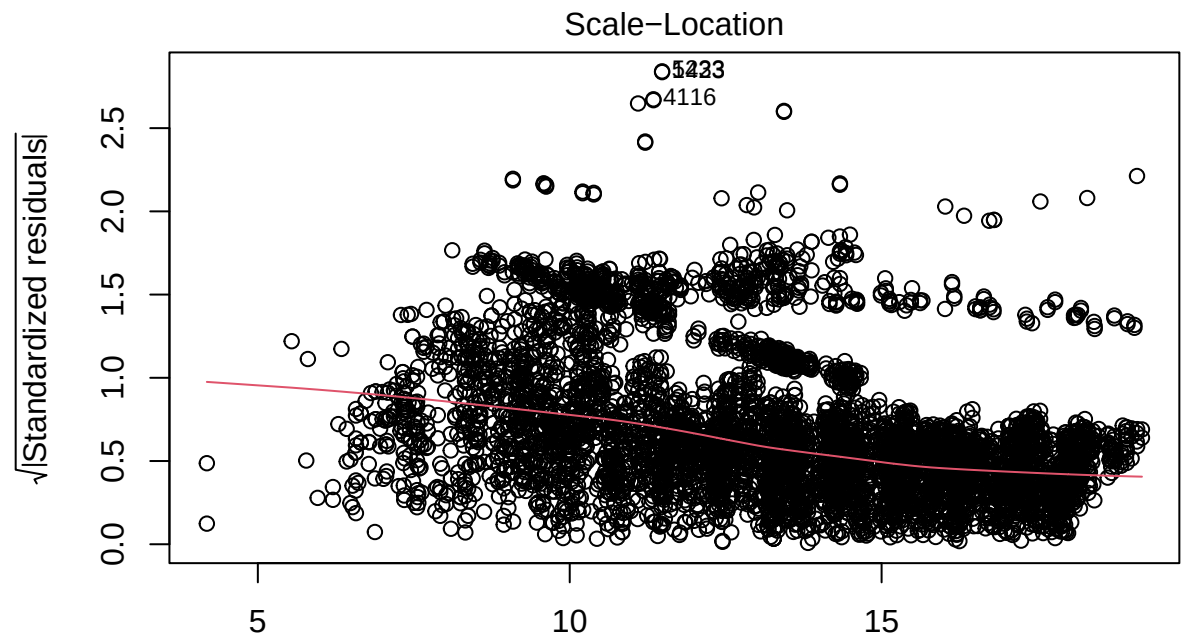
```
plot(mlr_enhanced)
```



Fitted values

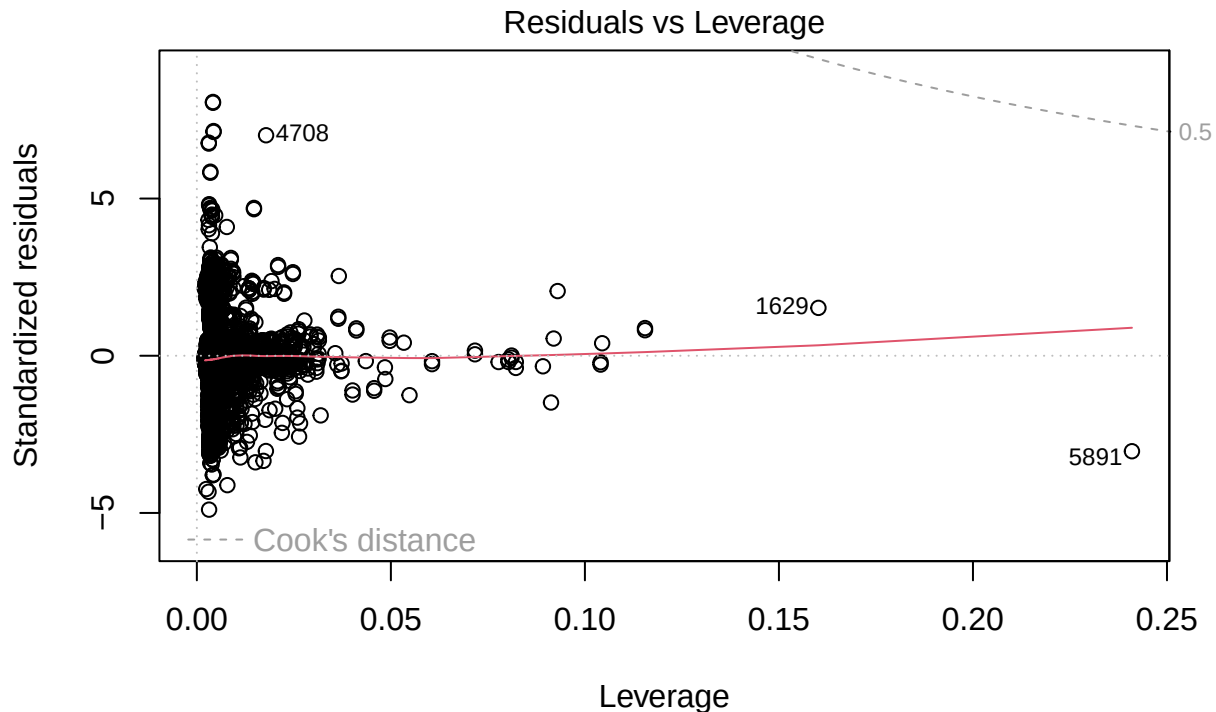
$\text{lm}(\text{sqrt_days} \sim \text{Breed_Cluster} + \text{Sex} + \text{Spayed_Neutered} + \text{Intake_Month} + \text{Age_a})$





Fitted values

$\text{lm}(\text{sqrt_days} \sim \text{Breed_Cluster} + \text{Sex} + \text{Spayed_Neutered} + \text{Intake_Month} + \text{Age_a})$



$\text{lm}(\text{sqrt_days} \sim \text{Breed_Cluster} + \text{Sex} + \text{Spayed_Neutered} + \text{Intake_Month} + \text{Age_a} .$

```
# VIF
```

```
print(vif(mlr_enhanced, type = "predictor"))
```

```
## GVIFs computed for predictors
```

```
##          GVIF Df GVIF^(1/(2*Df))          Interacts With
## Breed_Cluster  1.197235 18      1.005013          Age_at_Intake
## Sex            1.468443  2      1.100815          --
## Spayed_Neutered 1.493372  2      1.105457          --
## Intake_Month    1.169902 11      1.007158          --
## Age_at_Intake   1.197235 18      1.005013 I(Age_at_Intake^2), Breed_Cluster
##                                     Other Predictors
## Breed_Cluster          Sex, Spayed_Neutered, Intake_Month
## Sex                    Breed_Cluster, Spayed_Neutered, Intake_Month, Age_at_Intake
## Spayed_Neutered        Breed_Cluster, Sex, Intake_Month, Age_at_Intake
## Intake_Month            Breed_Cluster, Sex, Spayed_Neutered, Age_at_Intake
## Age_at_Intake          Sex, Spayed_Neutered, Intake_Month
```

```
# LASSO
```

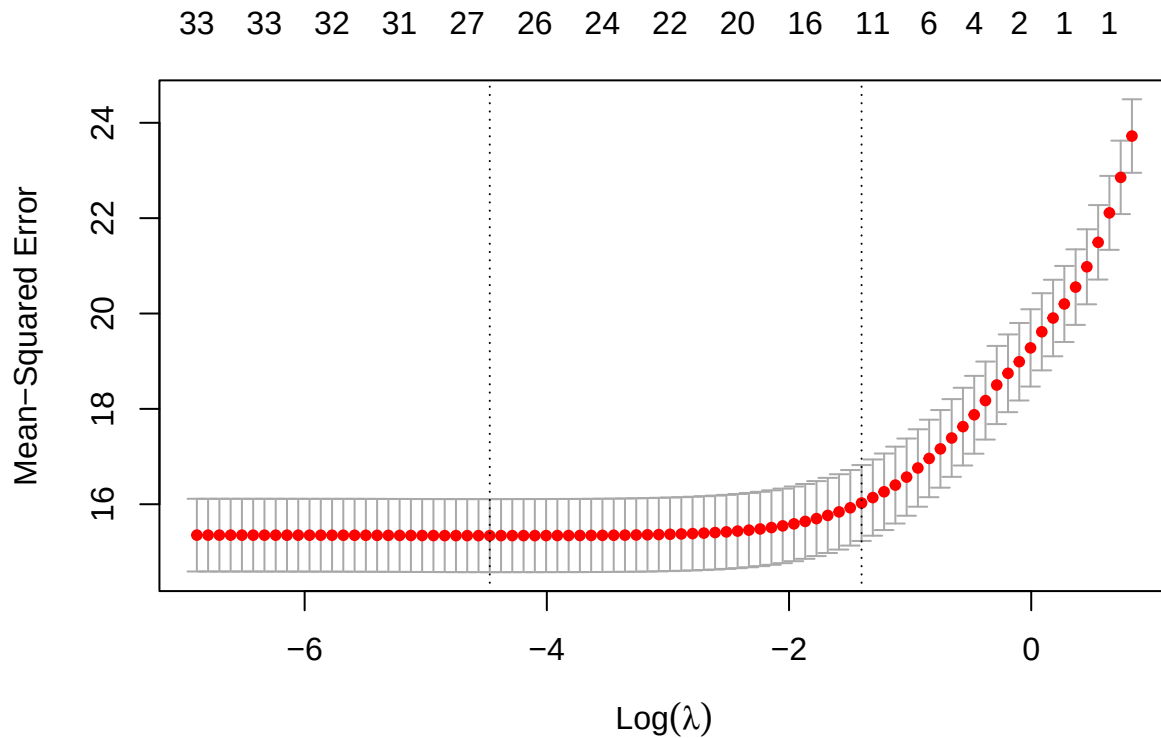
```
formula_enhanced <- as.formula(sqrt_days ~ Breed_Cluster + Sex + Spayed_Neutered + Intake_Month +
                                Age_at_Intake + I(Age_at_Intake^2) + Age_at_Intake:Breed_Cluster)
```

```
x_train <- model.matrix(formula_enhanced, train_data)[, -1]
```

```
y_train <- train_data$sqrt_days
```

```
x_test  <- model.matrix(formula_enhanced, test_data)[, -1]
```

```
cv_lasso <- cv.glmnet(x_train, y_train, alpha = 1)
plot(cv_lasso) # LASSO CV Plot
```



```
lasso_enhanced <- glmnet(x_train, y_train, alpha = 1, lambda = cv_lasso$lambda.min)
coef(cv_lasso, s = "lambda.min")
```

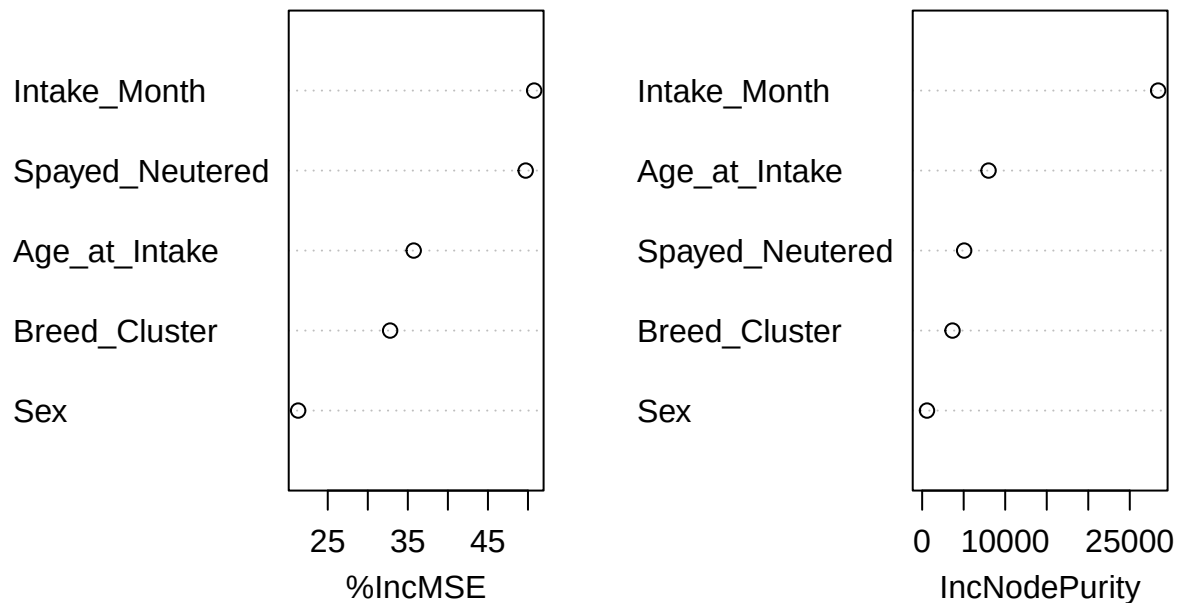
```
## 34 x 1 sparse Matrix of class "dgCMatrix"
##                                     s1
## (Intercept)                       11.944730732
## Breed_ClusterChihuahua Group      -1.066770902
## Breed_ClusterHound Group           .
## Breed_ClusterHusky Group           .
## Breed_ClusterOther/Small Mix      -0.160003455
## Breed_ClusterPit Bull Group        0.532657110
## Breed_ClusterRetriever Group       .
## Breed_ClusterShepherd Group       -0.124916183
## Breed_ClusterTerrier Group        -0.591019536
## SexMale                           0.054860690
## SexUnknown                        -0.052878130
## Spayed_NeuteredUnknown             0.046082746
## Spayed_NeuteredYes                 1.709263563
## Intake_Month.L                     2.469424451
## Intake_Month.Q                    -7.550586411
## Intake_Month.C                     0.165649647
```

```
## Intake_Month^4          3.670103776
## Intake_Month^5         -2.399399884
## Intake_Month^6         -0.244137340
## Intake_Month^7          1.402109185
## Intake_Month^8         -0.600125138
## Intake_Month^9         -0.368607461
## Intake_Month^10         0.804288523
## Intake_Month^11        -0.533251277
## Age_at_Intake           .
## I(Age_at_Intake^2)     -0.006524966
## Breed_ClusterChihuahua Group:Age_at_Intake .
## Breed_ClusterHound Group:Age_at_Intake .
## Breed_ClusterHusky Group:Age_at_Intake -0.207034222
## Breed_ClusterOther/Small Mix:Age_at_Intake -0.099309769
## Breed_ClusterPit Bull Group:Age_at_Intake 0.102443387
## Breed_ClusterRetriever Group:Age_at_Intake 0.070634755
## Breed_ClusterShepherd Group:Age_at_Intake -0.003450545
## Breed_ClusterTerrier Group:Age_at_Intake -0.061640759
```

```
# Random Forest
```

```
rf_enhanced <- randomForest(sqrt_days ~ Breed_Cluster + Sex + Spayed_Neutered +
                             Age_at_Intake + Intake_Month,
                             data = train_data, ntree = 500, importance = TRUE)
varImpPlot(rf_enhanced) # RF Variable Importance Plot
```

rf_enhanced



```

# Comparison
mlr_base_preds      <- predict(mlr_base, newdata = test_data)
mlr_enhanced_preds  <- predict(mlr_enhanced, newdata = test_data)
lasso_preds         <- as.vector(predict(lasso_enhanced, newx = x_test))
rf_preds            <- predict(rf_enhanced, newdata = test_data)

rmse <- function(actual, predicted) { sqrt(mean((actual - predicted)^2)) }

performance_table <- data.frame(
  Model = c("Base MLR", "Enhanced MLR", "Enhanced LASSO", "Enhanced Random Forest"),
  RMSE = c(rmse(test_data$sqrt_days, mlr_base_preds),
           rmse(test_data$sqrt_days, mlr_enhanced_preds),
           rmse(test_data$sqrt_days, lasso_preds),
           rmse(test_data$sqrt_days, rf_preds)),
  Adj_R2 = c(summary(mlr_base)$adj.r.squared, summary(mlr_enhanced)$adj.r.squared, NA, NA)
)
print(performance_table)

```

```

##           Model      RMSE      Adj_R2
## 1      Base MLR 4.901763 0.05975051
## 2    Enhanced MLR 4.043909 0.35712010
## 3    Enhanced LASSO 4.046677      NA
## 4 Enhanced Random Forest 3.935711      NA

```

```
summary(mlr_enhanced)
```

```

##
## Call:
## lm(formula = sqrt_days ~ Breed_Cluster + Sex + Spayed_Neutered +
##      Intake_Month + Age_at_Intake + I(Age_at_Intake^2) + Age_at_Intake:Breed_Cluster,
##      data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.0943  -1.4258  -0.4850   0.8233  31.4725
##
## Coefficients:
##              Estimate Std. Error t value
## (Intercept)      12.476556   0.601243  20.751
## Breed_ClusterChihuahua Group      -1.812965   0.637057  -2.846
## Breed_ClusterHound Group        -0.660406   0.871616  -0.758
## Breed_ClusterHusky Group        -0.456710   0.722136  -0.632
## Breed_ClusterOther/Small Mix    -0.794499   0.601763  -1.320
## Breed_ClusterPit Bull Group     -0.068011   0.610699  -0.111
## Breed_ClusterRetriever Group    -0.612258   0.609395  -1.005
## Breed_ClusterShepherd Group    -0.765571   0.619034  -1.237
## Breed_ClusterTerrier Group     -1.238068   0.630118  -1.965
## SexMale              0.077162   0.101574   0.760
## SexUnknown          -0.183971   0.415457  -0.443
## Spayed_NeuteredUnknown      0.172000   0.290532   0.592
## Spayed_NeuteredYes         1.748027   0.122955  14.217
## Intake_Month.L          2.542579   0.192368  13.217

```

## Intake_Month.Q	-7.593297	0.187040	-40.597
## Intake_Month.C	0.196107	0.186896	1.049
## Intake_Month^4	3.710659	0.182979	20.279
## Intake_Month^5	-2.424387	0.179228	-13.527
## Intake_Month^6	-0.292643	0.180598	-1.620
## Intake_Month^7	1.448105	0.175252	8.263
## Intake_Month^8	-0.643218	0.174881	-3.678
## Intake_Month^9	-0.411549	0.174772	-2.355
## Intake_Month^10	0.840070	0.165902	5.064
## Intake_Month^11	-0.578209	0.156929	-3.685
## Age_at_Intake	0.011795	0.148509	0.079
## I(Age_at_Intake^2)	-0.009711	0.003934	-2.468
## Breed_ClusterChihuahua Group:Age_at_Intake	0.044905	0.151122	0.297
## Breed_ClusterHound Group:Age_at_Intake	0.053628	0.224351	0.239
## Breed_ClusterHusky Group:Age_at_Intake	-0.253412	0.203140	-1.247
## Breed_ClusterOther/Small Mix:Age_at_Intake	-0.077865	0.146994	-0.530
## Breed_ClusterPit Bull Group:Age_at_Intake	0.126858	0.150452	0.843
## Breed_ClusterRetriever Group:Age_at_Intake	0.100689	0.151818	0.663
## Breed_ClusterShepherd Group:Age_at_Intake	0.011484	0.154715	0.074
## Breed_ClusterTerrier Group:Age_at_Intake	-0.042109	0.152376	-0.276
##	Pr(> t)		
## (Intercept)	< 2e-16	***	
## Breed_ClusterChihuahua Group	0.004444	**	
## Breed_ClusterHound Group	0.448672		
## Breed_ClusterHusky Group	0.527121		
## Breed_ClusterOther/Small Mix	0.186789		
## Breed_ClusterPit Bull Group	0.911330		
## Breed_ClusterRetriever Group	0.315082		
## Breed_ClusterShepherd Group	0.216239		
## Breed_ClusterTerrier Group	0.049481	*	
## SexMale	0.447486		
## SexUnknown	0.657915		
## Spayed_NeutererUnknown	0.553861		
## Spayed_NeutererYes	< 2e-16	***	
## Intake_Month.L	< 2e-16	***	
## Intake_Month.Q	< 2e-16	***	
## Intake_Month.C	0.294089		
## Intake_Month^4	< 2e-16	***	
## Intake_Month^5	< 2e-16	***	
## Intake_Month^6	0.105195		
## Intake_Month^7	< 2e-16	***	
## Intake_Month^8	0.000237	***	
## Intake_Month^9	0.018565	*	
## Intake_Month^10	4.23e-07	***	
## Intake_Month^11	0.000231	***	
## Age_at_Intake	0.936699		
## I(Age_at_Intake^2)	0.013603	*	
## Breed_ClusterChihuahua Group:Age_at_Intake	0.766367		
## Breed_ClusterHound Group:Age_at_Intake	0.811087		
## Breed_ClusterHusky Group:Age_at_Intake	0.212270		
## Breed_ClusterOther/Small Mix:Age_at_Intake	0.596328		
## Breed_ClusterPit Bull Group:Age_at_Intake	0.399160		
## Breed_ClusterRetriever Group:Age_at_Intake	0.507214		
## Breed_ClusterShepherd Group:Age_at_Intake	0.940835		

```
## Breed_ClusterTerrier Group:Age_at_Intake 0.782287
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.907 on 6122 degrees of freedom
## Multiple R-squared:  0.3606, Adjusted R-squared:  0.3571
## F-statistic: 104.6 on 33 and 6122 DF,  p-value: < 2.2e-16
```

```
rf_preds <- predict(rf_enhanced, newdata = test_data)

# Random Forest Residual vs. Fitter Plot
rf_residuals <- test_data$sqrt_days - rf_preds

rf_plot_data <- data.frame(Predicted = rf_preds, Residuals = rf_residuals)

ggplot(rf_plot_data, aes(x = Predicted, y = Residuals)) +
  geom_point(alpha = 0.4, color = "darkgreen") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  geom_smooth(method = "loess", color = "black", se = FALSE) + # Adds a trend line
  theme_minimal() +
  labs(title = "Random Forest: Residuals vs. Predicted Values",
       x = "Predicted Values (sqrt_days)",
       y = "Residuals")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

